HEXA Governance

The control and trust layer. Holds identity, tenant isolation, permissions, approvals and the tamper-evident record, so every mission stays inside the authority of the person who started it.

Who this is for: product, engineering, operations and anyone who has to explain XELOR to somebody else.
What it covers: the work HEXA reasons about, the rules it cannot break, the exact tools it can call, and what is honestly not built yet.
Truth standard: the repository as it stands on 6 August 2026.

HEXA

01 · AT A GLANCE

What HEXA is for

A role with a fixed tool list, not a general assistant

In one sentence

Holds identity, tenant isolation, permissions, approvals and the tamper-evident record, so every mission stays inside the authority of the person who started it.

3	2	4	0
business areas it speaks for	tools it can actually call	capability families allowed	agents it may delegate to

What reaches it

A verified token · The requested capability and permission · The proposed action plan · The recorded human decision

What it hands back

Allow or refuse, with the evidence · Preflight and post-execution verification · An approval requirement · A hash-chained, append-only trail

Capability families it is allowed to touch

`agent.` `workflow.` `governance.` `general.`

Ownership and authority are not the same thing

The areas above are where HEXA is the right specialist to ask. The tool table on page 8 is the much smaller set it can invoke at runtime. Owning a business area does not grant runtime power over it — and for ONYX, the supervisor, the gap between the two is the widest in the system.

What sits behind HEXA

Records, screens, workflow, controls and hand-offs

Organisation	HEXA module · general
Purpose	The legal and operating identity of a tenant: companies, registrations, departments and the document-number series shared by business modules.
Core records	Company master, legal name, base currency, GST registrations, registered addresses, departments and financial-year document sequences.
User surface	Companies; Departments. Company and registration APIs are also used by tax, numbering and agent context.
End-to-end flow	A tenant defines its company and registrations; operating modules resolve the correct seller identity and state; document services allocate gapless numbers inside the posting transaction.
Enforced controls	Tenant RLS, typed permissions, unique registration/document constraints and transaction-safe numbering. Tenant identity never comes from a browser-provided header.
Inputs and outputs	Feeds Sales tax, Purchase receiving, Accounts, statutory filings, module navigation and HEXA/ONYX company-context capabilities.
Current boundary	Company, registration, department and numbering foundations are live; it is not a full legal-entity consolidation or corporate-secretarial product.
Primary code map	`apps/web/src/modules/general/manifest.ts · apps/api/src/modules/general/`

Administration	HEXA module · administration
Purpose	The evidence and access centre for roles, separation of duties, security posture, incidents, privacy requests, audit integrity and licences.
Core records	139 permission keys, roles and grants, SoD rules/findings, access simulations, CERT-In incident clocks, privacy requests, audit-chain verification/anchors, retention and licence state.
User surface	Roles & access; Segregation; Security posture; Incidents; Privacy; Audit; Licence.
End-to-end flow	Define a role from catalogue permissions; grant it to a user; scan conflicting role pairs; enforce route guards at request time; append sensitive events; periodically re-hash the audit chain; review open statutory clocks and evidence packs.
Enforced controls	Catalogue trigger blocks unknown permissions, tenant RLS, server-side guards, named SoD rules, append-only audit, chain verification that identifies the break type and CI permission reconciliation.
Inputs and outputs	Every module consumes identity and permission decisions. Agent OS uses the same authority context; Integration and AI Operations report incidents and evidence into the control view.
Current boundary	Core access, SoD, audit and incident evidence are live. Row/field policy is mainly a preview, soft-revoked grants need a guard correction, and escalation/delegation automation is not complete.
Primary code map	`apps/web/src/modules/administration/manifest.ts · apps/api/src/modules/administration/`

How to read these module pages

“Live” means the records, service rules and user/API surface exist in this repository. It does not mean every surrounding enterprise process is complete. The boundary row names what remains illustrative, manual, simulate-driven or outside the MVP.

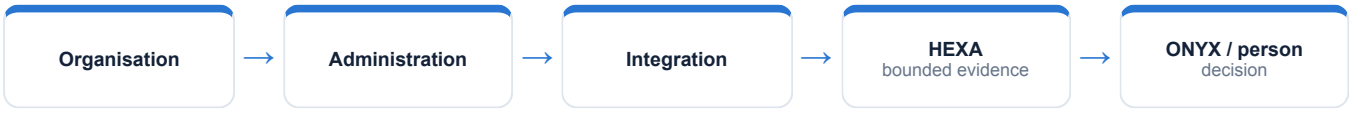
How HEXA's modules connect

A module owns its records; the agent explains the combined evidence

Integration	HEXA module · integration
Purpose	A governed place to describe external connections, trace messages and handle failures without pretending an unavailable downstream system is healthy.
Core records	Connector and connection definitions, flow state, message attempts, correlation traces, dead letters, circuit state, e-invoice/e-way-bill records and webhook secrets/deliveries.
User surface	Connections; Flows; Dead letters; Statutory filings; Webhooks.
End-to-end flow	Preflight a flow, classify each outcome, retry only transient failures with bounded policy, trip a circuit on repeated faults, quarantine terminal messages, and require a reviewed replay or resolution. Webhooks are signed and secrets can rotate.
Enforced controls	Idempotency/correlation ids, retry classification, circuit breakers, dead-letter custody, signed payload verification, secret rotation and statutory-window watches.
Inputs and outputs	Designed to connect Sales, Purchase, Accounts, statutory services and external consumers; feeds incidents and health evidence to Administration and Agent OS.
Current boundary	The resilience rules and screens are real and tested, but external network transports are simulate-driven in the MVP. No live GST, bank or supplier connector is claimed.
Primary code map	`apps/web/src/modules/integration/manifest.ts · apps/api/src/modules/integration/`

Cross-module coordination

Organisation supplies the legal and numbering context, Administration decides who may do what and proves the trail, and Integration carries external failures as explicit state. Together they are the control plane used by every business module and every agent run.



03 · THE WORK ITSELF

How HEXA's world actually runs

The business pipeline, not the agent plumbing

Four independent gates stand between a request and a row. Each is enforced by machinery rather than by wording, and three of the four are checked by CI on every build.

1 **Identity**

The tenant comes from a signature-verified Keycloak token's groups claim — never a request header. A proxy header is exactly the attack CVE-2025-29927 describes.

2 **Tenant fence**

withTenant() opens the transaction and sets app.current_tenant transaction-locally. Every tenant table has FORCE RLS; an unset tenant resolves to NULL and matches nothing.

3 **Permission**

139 keys shaped module.entity.action, typed at compile time. A permission that is not in the catalogue cannot even be granted — a trigger refuses the insert.

4 **Approval**

W1 resolves the approver from data — a named user or a role — and refuses anyone else by name with 403. The definition version is pinned, so publishing v2 never rewrites an approval in flight.

5 **Record**

Each audit row is sha256(previous hash + the canonicalised entry), sequenced per tenant. The append-only trigger fires for the schema owner too.

How much of this HEXA can actually see

Of everything above, HEXA can read exactly one thing directly — company masters. The rest of this pipeline is context for judging what it reads; it is not data the agent can reach. Where a mission needs more, it asks the specialist who owns it or it says the evidence is missing.

04 · THE RULES UNDERNEATH

What the code will not let anyone do

Enforced by constraints and locks, not by wording

app_user owns nothing and cannot bypass RLS

The API connects as a NOBYPASSRLS role that owns no table, so FORCE RLS is a real backstop rather than decoration.

The chain names how it broke

Verification distinguishes a deleted row (sequence gap), a replaced row (link mismatch) and an edited row (hash mismatch) — and stores the verdict, because “we verify nightly” is a claim and a dated row is evidence.

Ledger-critical writes never ride the bus

Stock, journals and budget commit synchronously in the caller's transaction. The outbox carries side effects only.

CI enforces the fence

db:rls-check asserts FORCE RLS and that a policy really keys on app.current_tenant; a two-tenant leak probe proves it live; perm-check reconciles all 139 routes against the registry.

Why this page matters more than the agent pages

An agent is only as trustworthy as the system it reads. Every rule here holds whether the request came from a person, a screen, a seeder or HEXA — which is precisely why an agent can be given a read tool without also being given a way to do damage.

Where these rules are kept

```
apps/api/src/common/tenant.middleware.ts
```

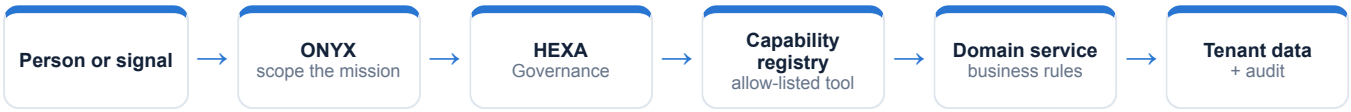
```
packages/platform/src/audit/hash-chain.ts
```

```
apps/api/src/modules/workflow/w1.service.ts
```

05 · RUNTIME CONTRACT

How HEXA takes part in a mission

The engine controls order, parallelism and recovery



1 · Scope

ONYX turns the goal into a run against a frozen graph version, with a fixed tenant, user, step budget and timeout.

2 · Authorise

Two checks, both required: the tool must be on HEXA's allow-list, and the signed-in person must independently hold the permission.

3 · Execute

A registered domain service does the work under the same rules a screen would face. There is no general SQL doorway.

4 · Preserve

Inputs, outputs, events and a checkpoint after every wave, so the run can be explained or resumed.

An agent can narrow authority, never widen it

HEXA is a specialist and delegates to nobody. Because the permission check is repeated against the requesting person, a mission can never reach data that person could not have opened themselves.

06 · EXACT CALLABLE SURFACE

Everything HEXA can call

This table is the complete list — there is no other door

Capability	Mode	What comes back	Permission required	Needs approval
general.companies.read	Read	Company masters	general.company.read	No
agent.action.dispatch	Execute	Governed work item	agentos.run.operate	Yes

Read · Analyse · Simulate

Return evidence or a reproducible view. Nothing in the business changes.

Draft

Produce reviewable content that is labelled a draft and can never present itself as an approved outcome.

Execute

The only side-effecting mode in the whole system, and the only one that requires an approved human gate above it.

Both locks must open

The agent's allow-list AND the person's permission. Either one failing is a refusal.

What “dispatch” does and does not do

It appends a governed work item naming the responsible specialist, the approval it came from and the exact payload. It does not change a sales order, a stock balance, a production order, a journal or a customer message — a person still does that work.

07 · IN THE SIX MISSIONS

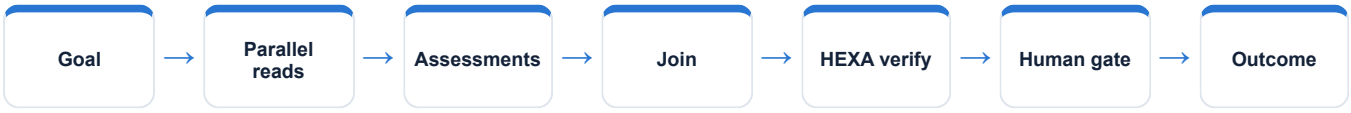
Where HEXA actually appears

Node by node, from the registered graph definitions

Mission graph	What HEXA does in it
Cross-functional readiness	Verifies policy and evidence before the human gate
Eight-agent operating review	Reads company context, assesses control, runs four checks
Controlled action mission	Context, assessment, preflight, one dispatch, outcome verification — 5 nodes
Working Capital Review	Verifies every figure ties to source and no posting occurred
QMS & Audit Readiness	Verifies traceability and that no AI decided a result
Managed Service Assurance	Verifies evidence, one-owner boundaries and reserved decisions

> **Reading this honestly**
>
> “Not involved” is not a limitation, it is the design. A mission asks only the specialists whose evidence it needs, and a specialist cannot add itself to a graph at runtime. The graph is written down, versioned and content-hashed before the run starts.

The shape every mission shares



08 · WORKED EXAMPLE

Can this persona open another department?

Real records from the running demo, not an illustration

Signed in as mica.commercial, the other six accountable departments are visibly shut — shown and dimmed, not hidden, because pretending they do not exist would redraw the company.

Typing /department/SPAR into the address bar does not help. The SERVER refuses: the same token asking the API for another department's data gets 403, resolved from grants in tenant-fenced tables.

Nothing in between could be edited in a browser's developer tools, because nothing in between is making the decision.

> **The sentence to say out loud**
> “HEXA contributes evidence from its own area, with the source records behind it and its limits stated. It does not claim an action is done until the system has recorded and verified it.”

Fair questions to ask while watching

- Which records is this answer standing on?
- Which permission was checked, and against whom?
- Is this a recorded fact, a simulation, a draft, or a completed result?
- Would this step have waited for a person?
- What happens if the service restarts halfway through?

09 · WHAT IS REAL

Live today, and deliberately not

The second column is the one worth reading

Working now

- FORCE RLS on every tenant table, CI-verified
- 139 typed permissions with a catalogue trigger
- W1 approvals with a hash-chained trail
- Audit verification, anchoring and the auditor pack
- Circuit breakers, DLQ triage and signed webhooks as pure, tested logic

Not built — stated plainly

- The permission guard does not filter is_active, so a soft-revoked grant is still honoured — a real divergence from the access simulator
- W1 has no cancel, no delegation and no escalation; the SLA is a pull query with no timer
- Row-scope and field masking are reachable only through the access preview, not applied by list endpoints
- Integration has no transport: the resilience logic is real and tested, the I/O half is simulate-driven

Identity boundary	Verified token → tenant and actor
Authority boundary	Agent allow-list AND the person's permission
Action boundary	Side effects need an approved ancestor node
Data boundary	Domain service over a tenant-fenced database
Evidence boundary	Append-only runs, events, checkpoints and audit

Gaps are listed because a guide that hides them fails the first hour of technical due diligence. Every item on the right is either a roadmap decision or a known defect, and both are recorded in the project's own capability-gap register.

10 · QUICK REFERENCE

HEXA on one page

For explaining the agent to somebody new

Its job

Holds identity, tenant isolation, permissions, approvals and the tamper-evident record, so every mission stays inside the authority of the person who started it.

Speaks for

Organisation & master data, Administration, Integration

Takes in

A verified token · The requested capability and permission · The proposed action plan · The recorded human decision

Gives back

Allow or refuse, with the evidence · Preflight and post-execution verification · An approval requirement · A hash-chained, append-only trail

Can call

general.companies.read, agent.action.dispatch

Cannot do

The permission guard does not filter is_active, so a soft-revoked grant is still honoured — a real divergence from the access simulator · W1 has no cancel, no delegation and no escalation; the SLA is a pull query with no timer · Row-scope and field masking are reachable only through the access preview, not applied by list endpoints

Words used on these pages

Agent	A named specialist with a fixed, allow-listed set of tools — not a process that can roam.
Capability	One registered operation with a declared mode, owner and required permission.
Mission graph	A versioned recipe: which steps run, which run together, where it waits for a person.
Checkpoint	A stored snapshot after each wave, used to explain a run and to resume it safely.
Governed work item	An approved, append-only assignment for a human team — not the business change itself.